

This event will begin shortly.

VIRTUAL EVENT

Deconstructing AI Capex: How Much More Growth Ahead?

September 30, 2025

Bloomberg Intelligence

500+

data contributors

135+

industries

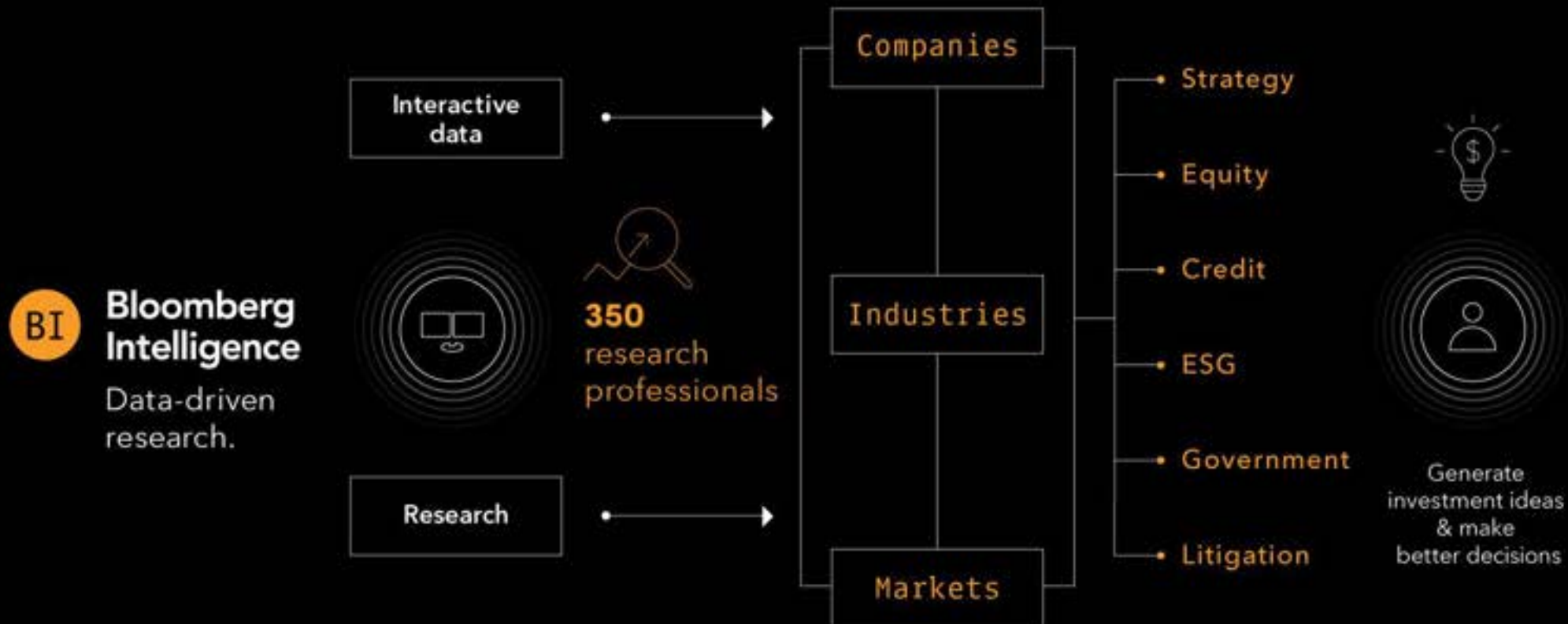
2,000+

companies

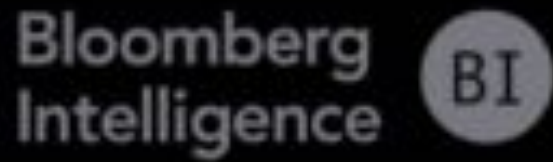
15yrs

avg. analyst experience

Bloomberg Intelligence (BI) research delivers an independent perspective providing interactive data and investment research on companies, industries and global markets. Our team of 350 research professionals help our clients make informed decisions in the rapidly moving investment landscape.



Moderators and Speakers



Woo Jin Ho

Senior Analyst
Hardware and Networking
Bloomberg Intelligence

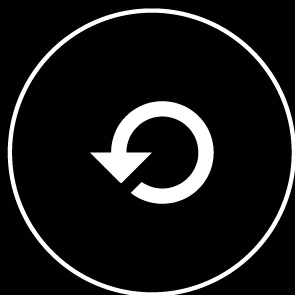


Jeremie Ontiveros

Analyst
SemiAnalysis

Administrative notes

- Today's presentation will be recorded and available for playback.
- Slides will be emailed to participants after the event.
- Ask a question by submitting one in the Q&A box.
- After the webinar do reach out to our analysts for further questions.
- Click on the icons shown below in the toolbar below to adjust windows:



Restores screens
to default setting



Restores
Q&A box

Agenda

- SemiAnalysis Overview
- Understanding the Coming CAPEX Surge
- Individual Hyperscaler Focus
- Economics and ROI of GPU Clouds
- SemiAnalysis Industry Models
- SemiAnalysis Core Research

SEMIANALYSIS | SECTOR COVERAGE & ANALYST TEAM

Sector Coverage

Subassemblies, Materials,
Consumables

Semicap Equipment

Testing and Packaging

Foundry and IDMs

IC Design

ODM and OEMs

Networking, Optical,
Datacenters

AI Companies, AI Models,
Cloud Services

Tokenomics, Applications,
Hardware

Analyst Team

Research

Dylan Patel – CEO, Founder, and Chief Analyst
Doug O’Laughlin – Strategy and Financial Research
Dan Nishball – Networking, Neoclouds, Data Centers
Myron Xie – Semicap, Advanced Packaging, Foundry
Jeff Koch – Wafer Fab Equipment, Process Technology
Jeremie Eliahou Ontiveros – Datacenters, Power, Analog
Wei Zhou – Tokenomics and AI Usage
Sravan Kundojjala – Foundry, Memory
Allison Elliot – Reverse Engineering
Andrew Wagner – Reverse Engineering
Dan Frech – Reverse Engineering
Daniel Sanchez – Reverse Engineering
Gerald Wong Chian Hao – VLSI, Design, PDK
Ivan Chiam – Packaging and Accelerator
Wega Chu – Server OEMs and ODMs, Components
Randy Chiang – Fabs, Semicap, Taiwan
Michael Chen – Foundry and Accelerators
David Chung – AI Supply Chain, Packaging
Tanj Bennett – Systems Architecture and Hardware
Ajey Pandey – Grid, Energy Research and Modelling
Maya Barkin – Datacenters and Power
Reyk Knuhtsen – Robotics, Datacenter and Power
Kang Wen Cheang – Neoclouds and Systems

Patrick Zhou – Networking and Optics

Clara Ee – Networking and Optics

Chaim Eisenberg – Chips and Wafers Data Product

Analyst Team (cont’d)

Research

Jordan Nanos – AI Engineer
Kimbo Chen – AI Engineer
AJ Kourabi – AI Models and Usage
Max Schmalzl – Multimedia Content Creation
Nick Doyle – Financial Research and Analysis



Operations and Sales Team

Operations

Michelle Shen – Chief of Staff
Chirag Mendpara – Finance and Operations
Emeline De Sousa – Customer Success
Kyle Johntz Connor – Tech / Data Lead, Data Science
Robert Paul Rice – IT Management
Mercy Swinney – Data Analyst

Sales

David Wolfson – Sales Manager and Operations
Andrew Lekashman – Sales and Operations

SEMIANALYSIS PRODUCT OVERVIEW

Accelerator Model

Datacenter Model

AI TCO Model

WFE Model

Datacenter Market
Map

BOM Cost Reports
GB200, Mi300,
Trainium2

GPU Pricing
Series

Financial Research
Product
(Core Research)

Optics and
Networking Model

Foundry Industry
Model

Tokenomics
Model

NEW

NEW

NEW

August 2025

UNDERSTANDING THE COMING CAPEX SURGE

OUR CAPEX FORECAST (PRE Q2 2025 EARNINGS)

Hyperscaler CapEx: over half a trillion USD by 2026

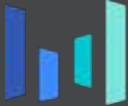
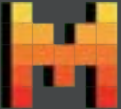
- 37 / 58% ahead of consensus in 2026 / 27
- Oracle's ramp remains widely underestimated: the \$30B/year OpenAI deal alone requires ~\$75B CapEx
- Methodology: a blend of our industry models
 - Tracking 5,000+ individual datacenters (SemiAnalysis Datacenter Model)
 - Hundreds of supply-chain datapoints (SemiAnalysis Accelerator Model)
 - Detailed hyperscaler-by-hyperscaler system and networking pricing (SemiAnalysis AI Cloud TCO model, AI Networking Model)

	Unit	2023	2024	2025	2026	2027
Hyperscaler CapEx - SemiAnalysis vs Consensus						
Microsoft						
SemiAnalysis - Microsoft	USD Million	\$ 35,202	\$ 55,552	\$ 72,830	\$ 97,536	\$ 111,130
Consensus	USD Million	\$ (35,202)	\$ (55,552)	\$ (69,016)	\$ (76,526)	\$ (83,976)
% Difference	%	0%	0%	6%	27%	32%
Meta						
SemiAnalysis - Meta	USD Million	\$ 27,266	\$ 37,256	\$ 70,421	\$ 94,930	\$ 113,557
Consensus	USD Million	\$ (27,266)	\$ (37,256)	\$ (67,555)	\$ (78,577)	\$ (77,675)
% Difference	%	0%	0%	4%	21%	46%
Google						
SemiAnalysis - Google	USD Million	\$ 32,251	\$ 52,535	\$ 75,753	\$ 107,524	\$ 140,656
Consensus	USD Million	\$ (32,251)	\$ (52,535)	\$ (73,244)	\$ (77,774)	\$ (82,605)
% Difference	%	0%	0%	3%	38%	70%
Amazon						
SemiAnalysis - Amazon	USD Million	\$ 52,372	\$ 82,145	\$ 105,643	\$ 132,005	\$ 138,068
Consensus	USD Million	\$ (52,372)	\$ (82,145)	\$ (104,261)	\$ (109,237)	\$ (113,026)
% Difference	%	0%	0%	1%	21%	22%
Oracle						
SemiAnalysis - Oracle	USD Million	\$ 6,617	\$ 10,108	\$ 26,927	\$ 41,444	\$ 74,932
Consensus	USD Million	\$ (6,617)	\$ (10,108)	\$ (23,675)	\$ (27,301)	\$ (29,382)
% Difference	%	0%	0%	14%	52%	155%
Coreweave						
SemiAnalysis - Coreweave	USD Million	\$ 1,224	\$ 7,380	\$ 25,406	\$ 33,548	\$ 33,630
Consensus	USD Million	\$ (1,224)	\$ (7,380)	\$ (21,500)	\$ (15,246)	\$ (13,400)
% Difference	%	0%	0%	18%	120%	151%
Hyperscaler Total						
SemiAnalysis - Total	USD Million	\$ 154,932	\$ 244,976	\$ 376,979	\$ 506,987	\$ 611,972
Consensus	USD Million	\$ (153,708)	\$ (237,596)	\$ (337,751)	\$ (369,415)	\$ (386,664)
% Difference	%	1%	3%	12%	37%	58%
Y/y Growth - SemiAnalysis, %	%		58%	54%	34%	21%
Y/y Growth - Consensus, %	%		55%	42%	9%	5%

Source: sales@semianalysis.com

WHAT DRIVES THE >\$0.5T HYPERSCALER CAPEX?

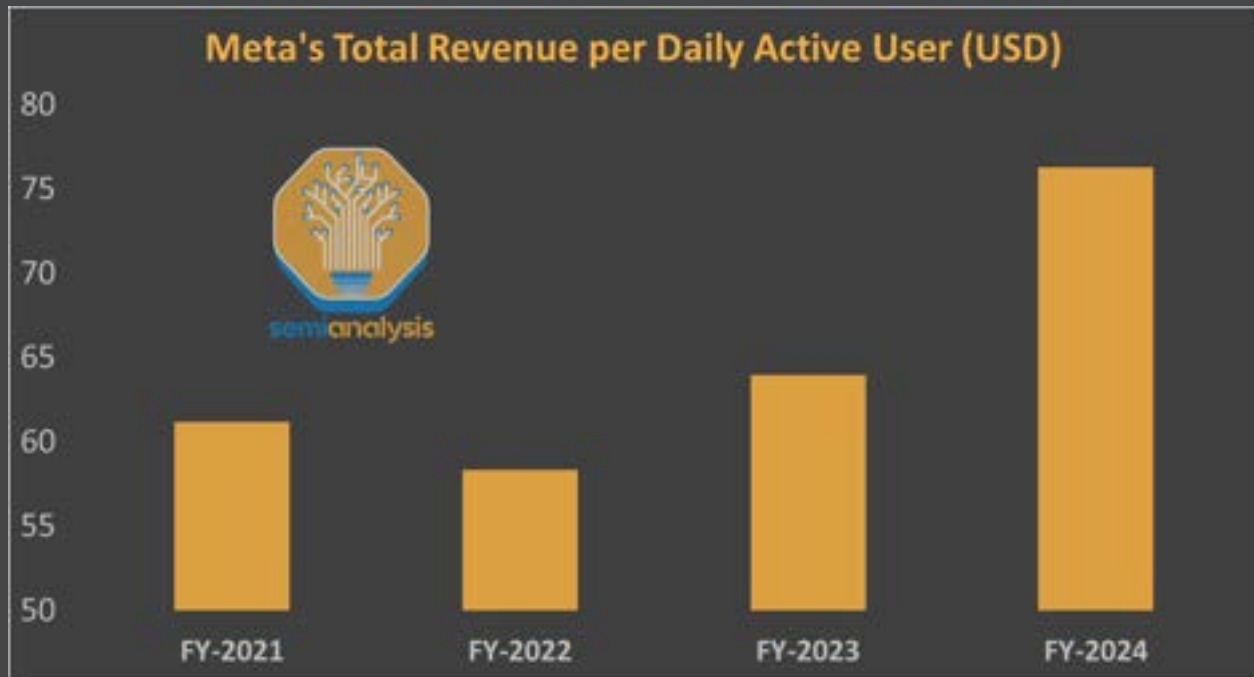
Three main sources of demand for GPUs (and associated infra such as Datacenters):

- 1/ Advertising  Meta  ByteDance  Google
- 2/ Frontier AI Labs  OpenAI  AI  X1  DeepMind  LLaMA by  Meta
- 3/ Everything else (startups, enterprise,...)  Jane Street  MISTRAL AI  deepseek  perplexity

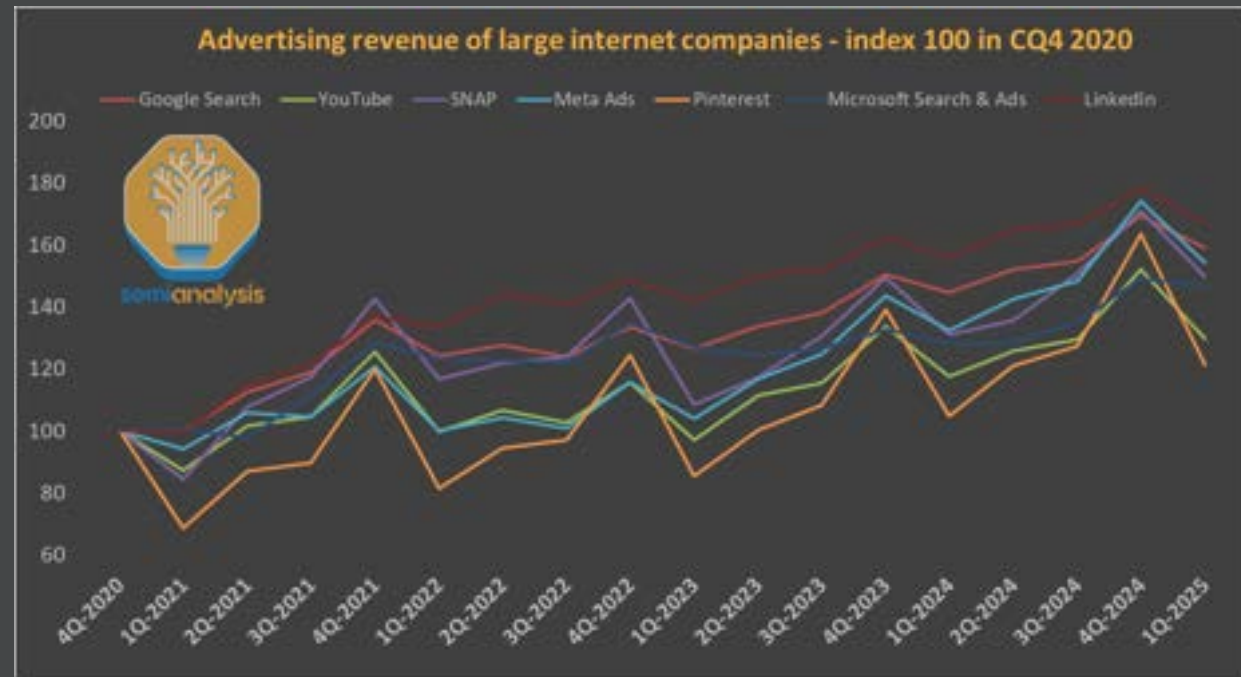
1/ ADVERTISING

High quality, high-ROI GPU investments:

- Meta and Google are among the fastest-growing social media + advertising firms ...
- ...Despite a highly mature, stalling user base. **Monetization/user is surging**
- Ad-related GPUs and TPUs investments are surging: as per Meta Q1'25 earnings call, “over half of our CapEx goes to the Core Business”)



Source: Company filings

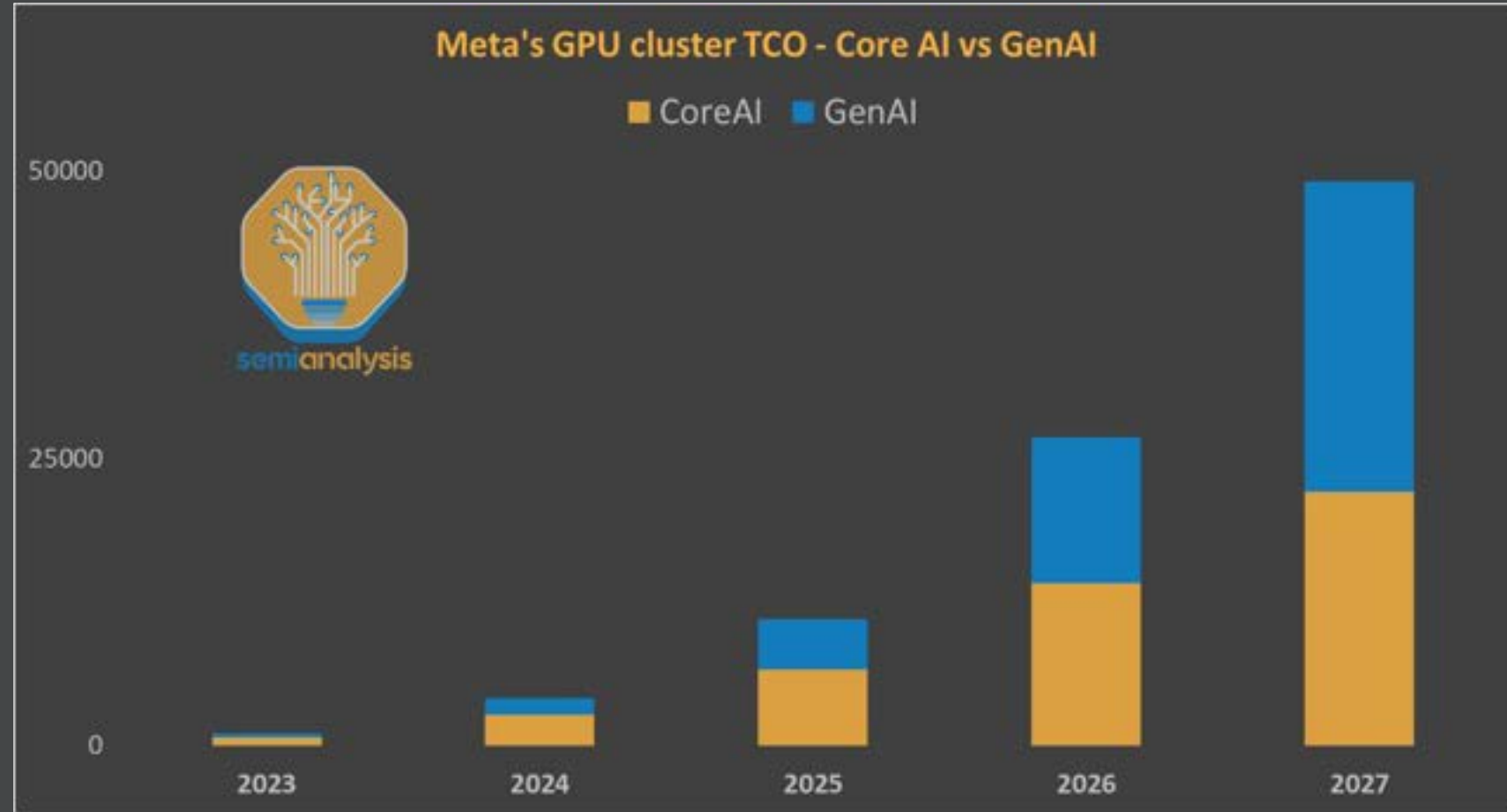


Source: Company filings

1/ ADVERTISING

Meta's \$25B TCO by 2027 on "Core AI"-related GPUs:

- Assuming a 10% revenue growth contribution per year in 2024-27
- "Core AI" adds >\$50B of revenue, on \$25B TCO
- This includes a substantial portion of "Core AI" GPUs for Training, i.e. inference margin is even higher
- We think **Google** and **ByteDance** are applying the same playbook



Source: sales@semianalysis.com

2/ FRONTIER AI LABS

OpenAI, Anthropic, xAI CapEx needs weigh ~20% of hyperscaler CapEx:

- We estimate the AI Lab capacity requirements by
 - 1/ Identifying their large AI Training datacenters
 - 2/ Assuming a margin on inference to derive Cloud Spending and required capacity
- Serving the three big independent AI labs in 2026 will require north of \$100B of CapEx

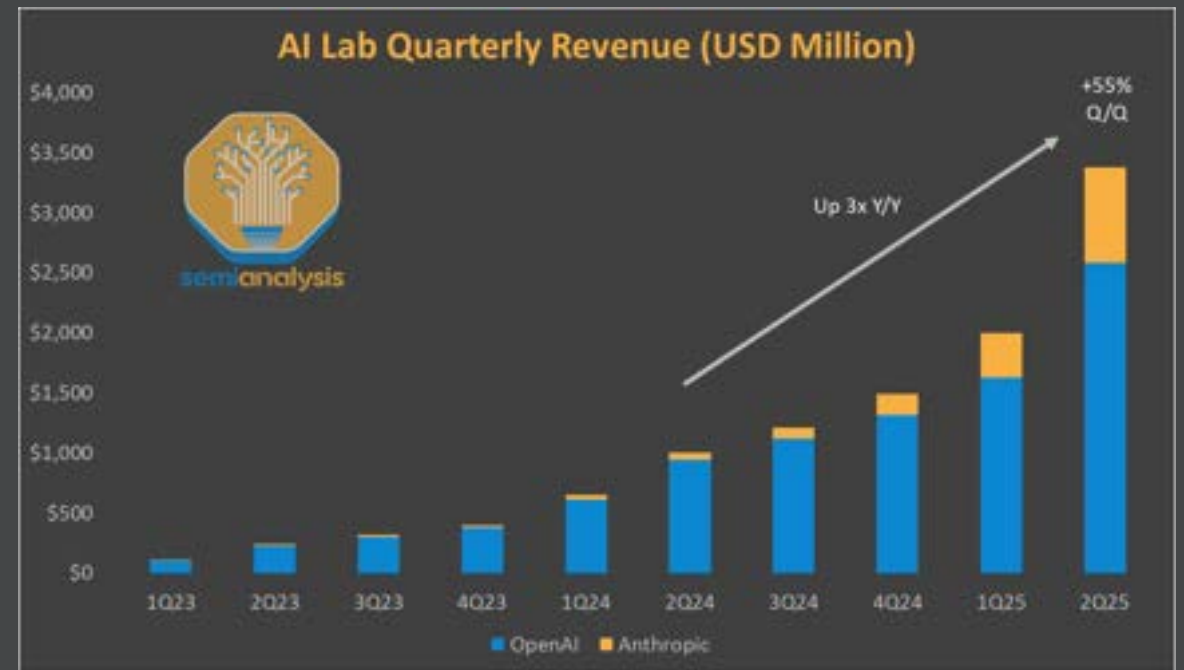
	Unit	2023	2024	2025	2026	2027
SemiAnalysis AI Lab Capex & Compute Forecast						
OpenAI - MW Added	MW	159	559	1,058	1,808	2,005
xAI- MW Added	MW	31	166	414	484	328
Anthropic- MW Added	MW	36	166	972	1,163	1,596
OpenAI - CapEx per MW (DC + XPU Cluster)	USD mm Per MW	35	35	35	36	36
xAI - CapEx per MW (DC + XPU Cluster)	USD mm Per MW	33	33	33	33	33
Anthropic - CapEx per MW (DC + XPU Cluster)	USD mm Per MW	22	22	22	22	22
OpenAI-related CapEx	USD Million	5,569	19,680	37,417	64,236	71,587
xAI-related CapEx	USD Million	998	5,408	13,602	15,971	10,890
Anthropic-related CapEx	USD Million	791	3,659	21,593	25,978	35,829
Total	USD Million	7,358	28,747	72,612	106,185	118,307
As a % of Hyperscaler CapEx		5%	12%	19%	21%	20%

Source: sales@semianalysis.com

2/ FRONTIER AI LABS

The spending rests on aggressive future ARR targets:

- Anthropic and OpenAI are both looking to generate tens of billions of ARR by 2026...
- ... But the aggressive forecast is backed by impressive growth since Q1'23.
- In 2025 YTD, OpenAI is on track with expectations, while Anthropic is materially ahead

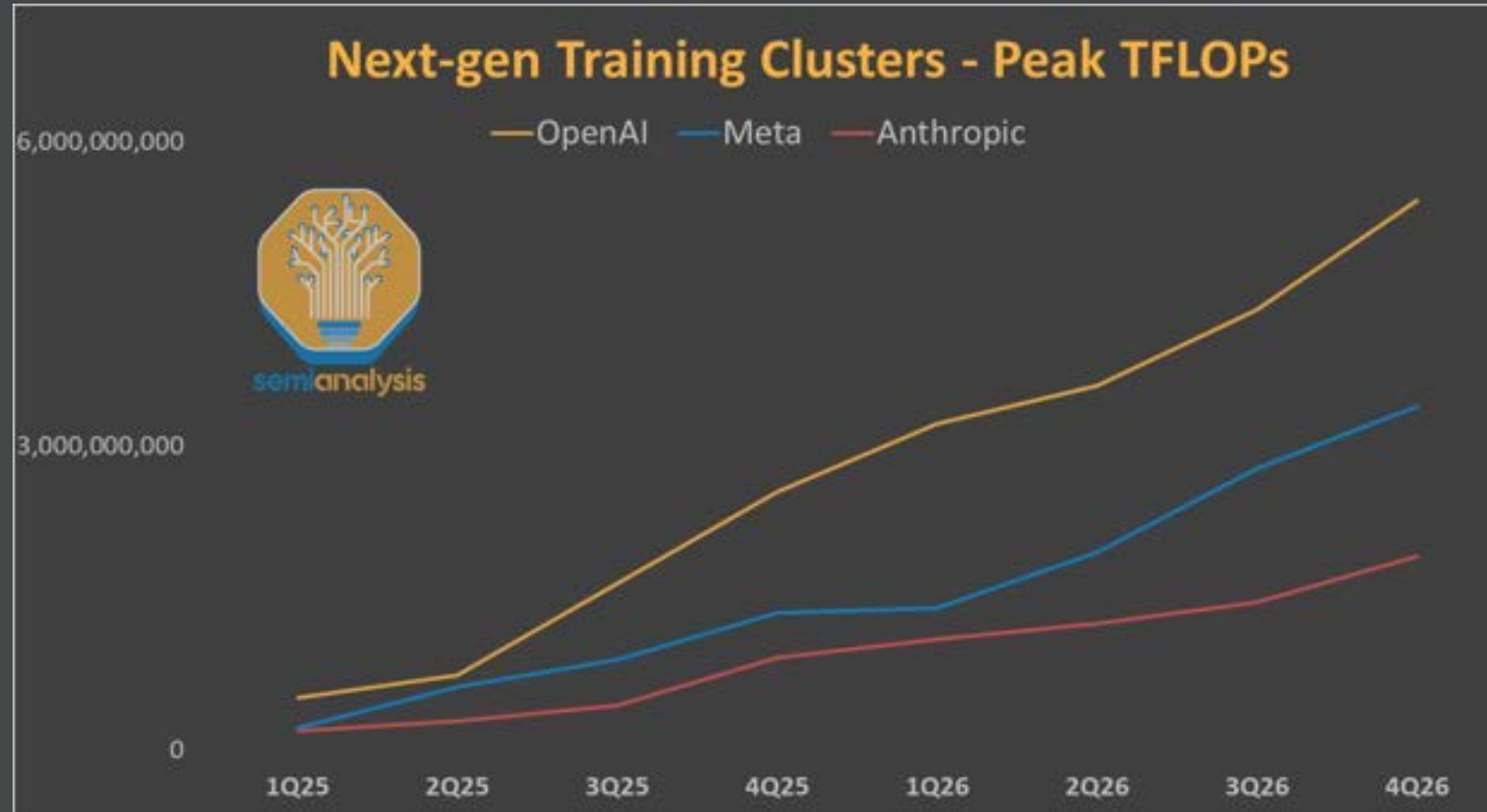


Source: Reuters, The Information, Bloomberg

2/ FRONTIER AI LABS

Training remains a driving force:

- Training-related spending will surge in H2'25, 2026 and 2027, as large AI Training Datacenters turn on
- All frontier AI labs have **multiple multi-hundred-MW** AI projects

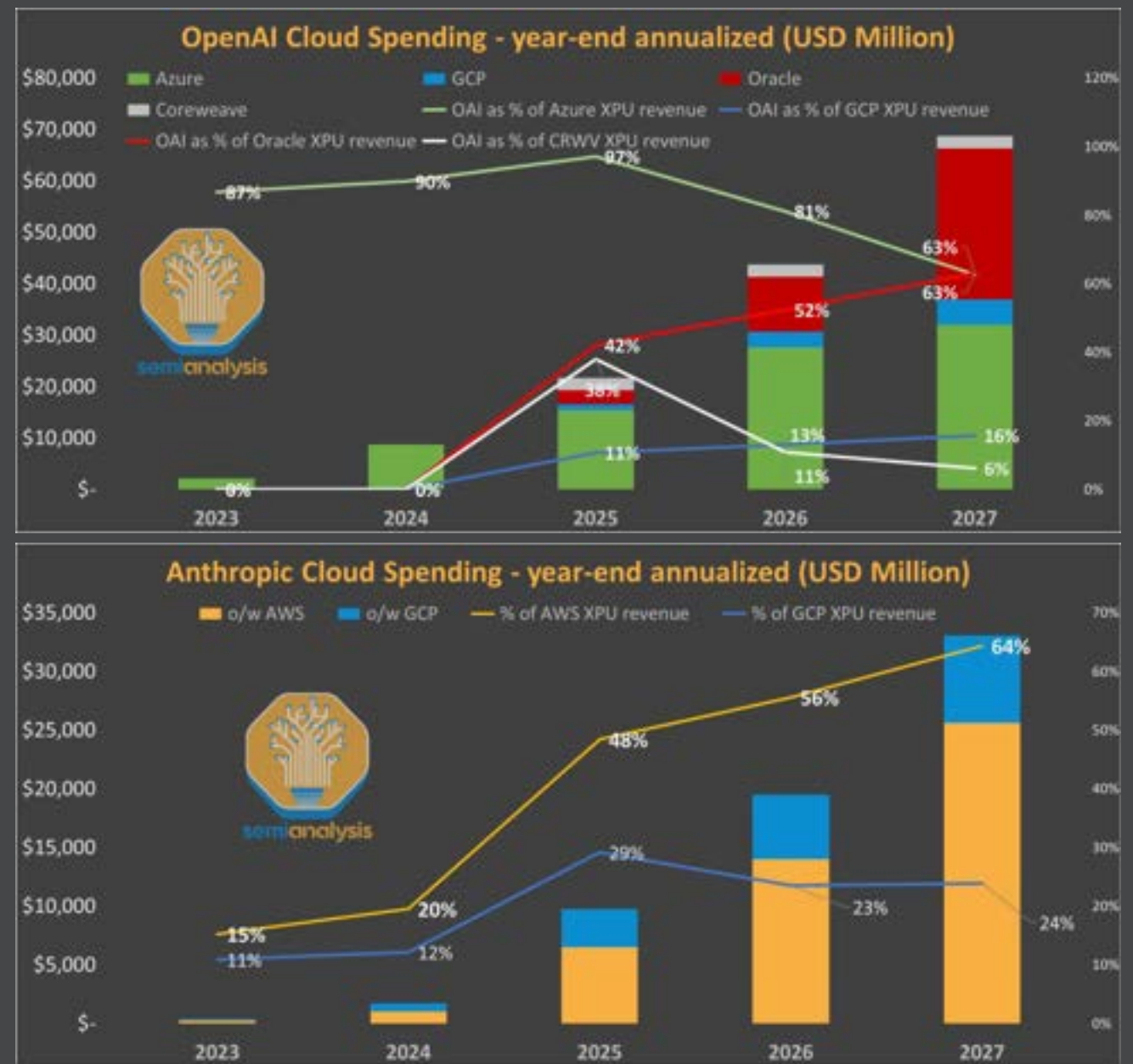


Source: sales@semianalysis.com

2/ FRONTIER AI LABS

AI Lab spending drives growth for Clouds:

- We think the majority of Azure's outperformance is driven by OpenAI (100% market share to date, will change in H2'25)
- AWS will strongly benefit from Anthropic's **training** ramp, which hits Q3/Q4'25, while GCP captures more inference



Source: sales@semianalysis.com

3/ EVERYONE ELSE

A harder-to-track, highly heterogenous category

- 1/ **Startups**: includes mid-level AI labs (Mistral AI, Cohere, SSI, Thinking Machines...), applications trying to go vertical (Cursor, Perplexity, Windsurf...), and many small startups
 - These companies sit in a highly competitive market, and are highly dependent upon market funding conditions
- 2/ **Enterprises**: financial firms, automotive, client service, healthcare, business process automation...
 - Different levels of maturity but overall early stage/ "POC" phase.

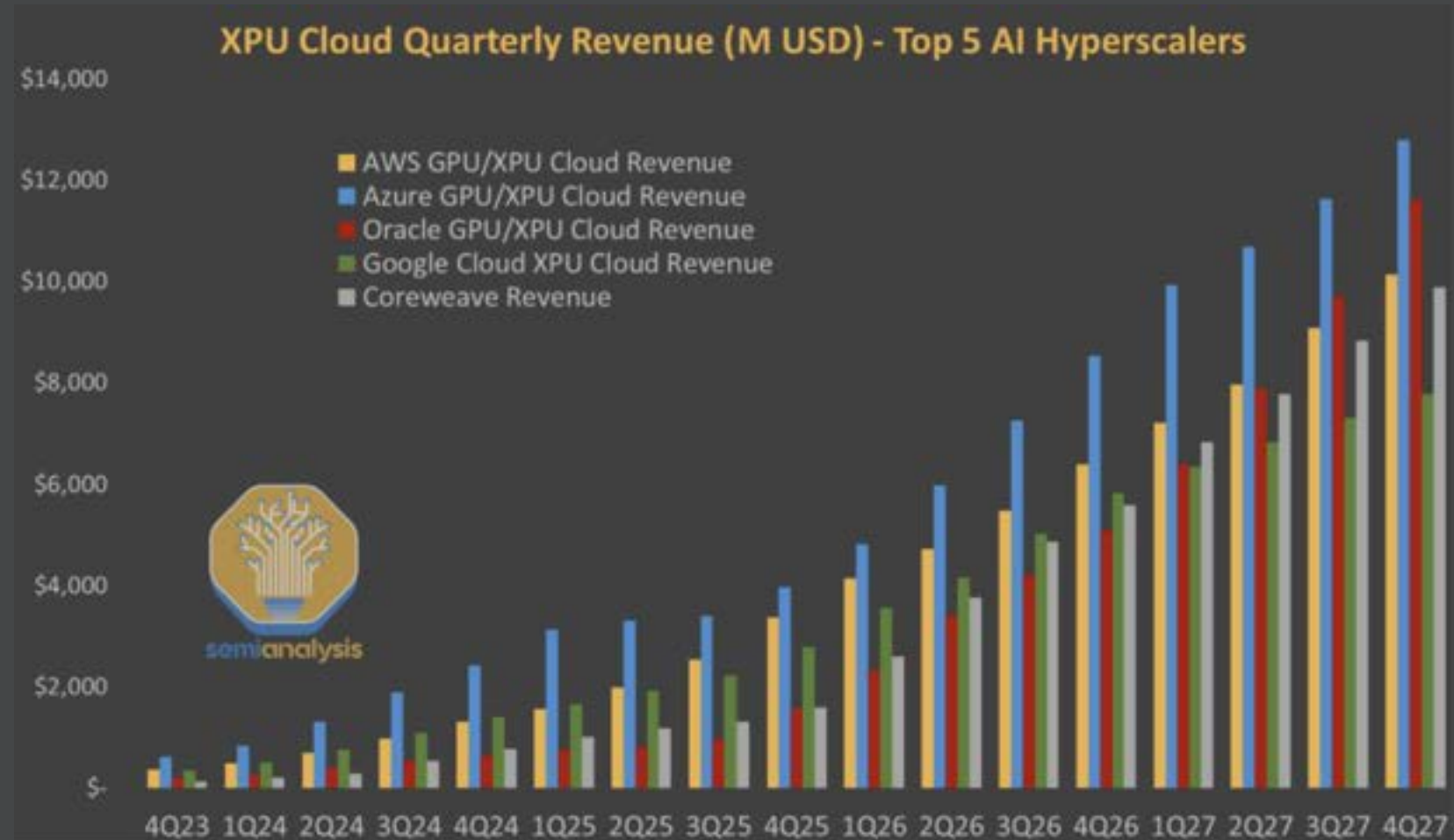
August 2025

INDIVIDUAL HYPERSCALER FOCUS

CLOUD REVENUES TO ACCELERATE

Our bottom-up GPU/XPU Cloud Revenue model shows strong growth ahead

- We expect a strong revenue acceleration in late 2025 and 2026 for most Clouds
- The top 5 largest Clouds will surpass \$5B of quarterly XPU revenue by Q4'26
- Large AI deals drive significantly more growth than enterprise/startup

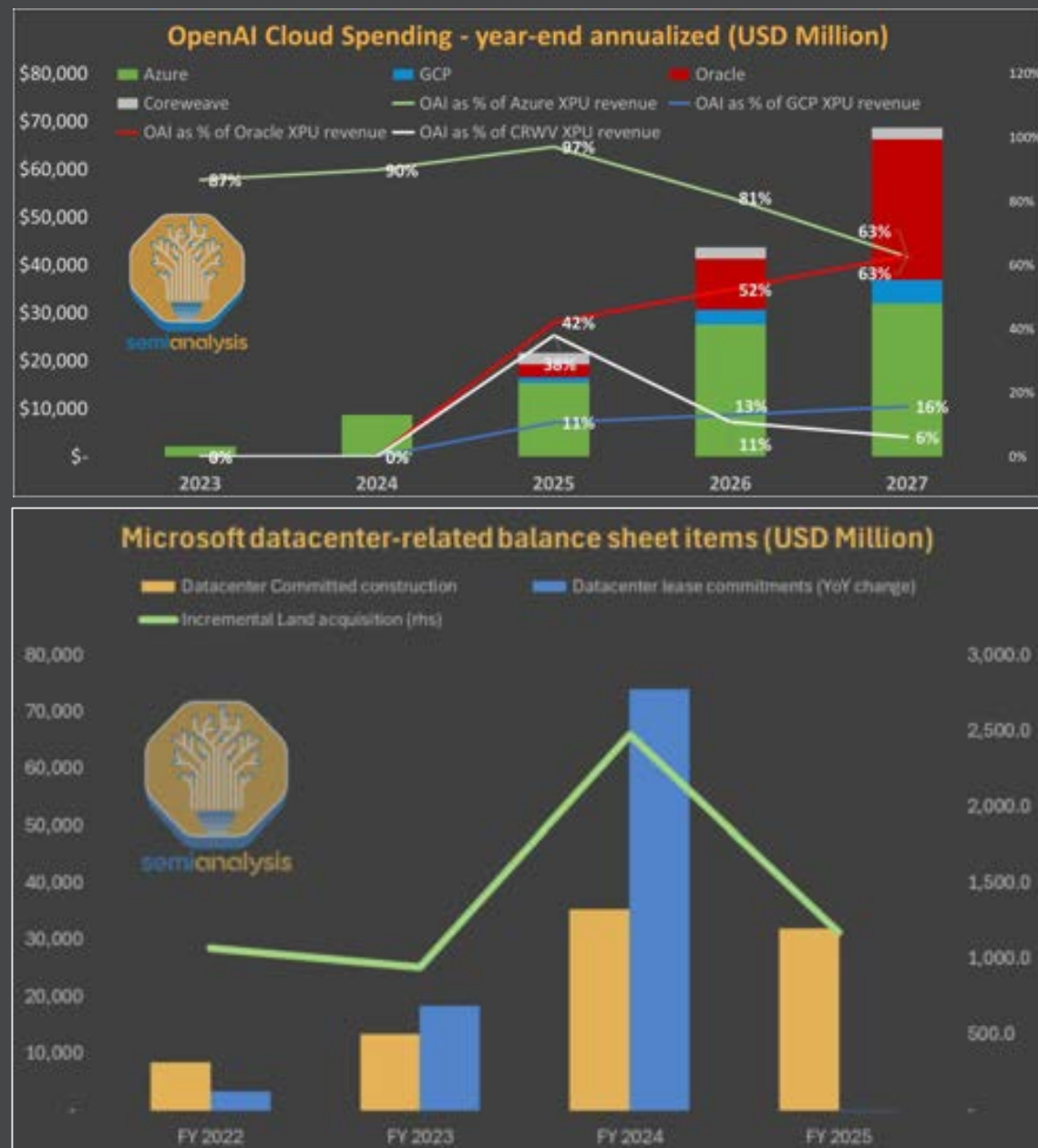


Source: sales@semianalysis.com

MICROSOFT

Strong 2025-26, uncertainty in 2027:

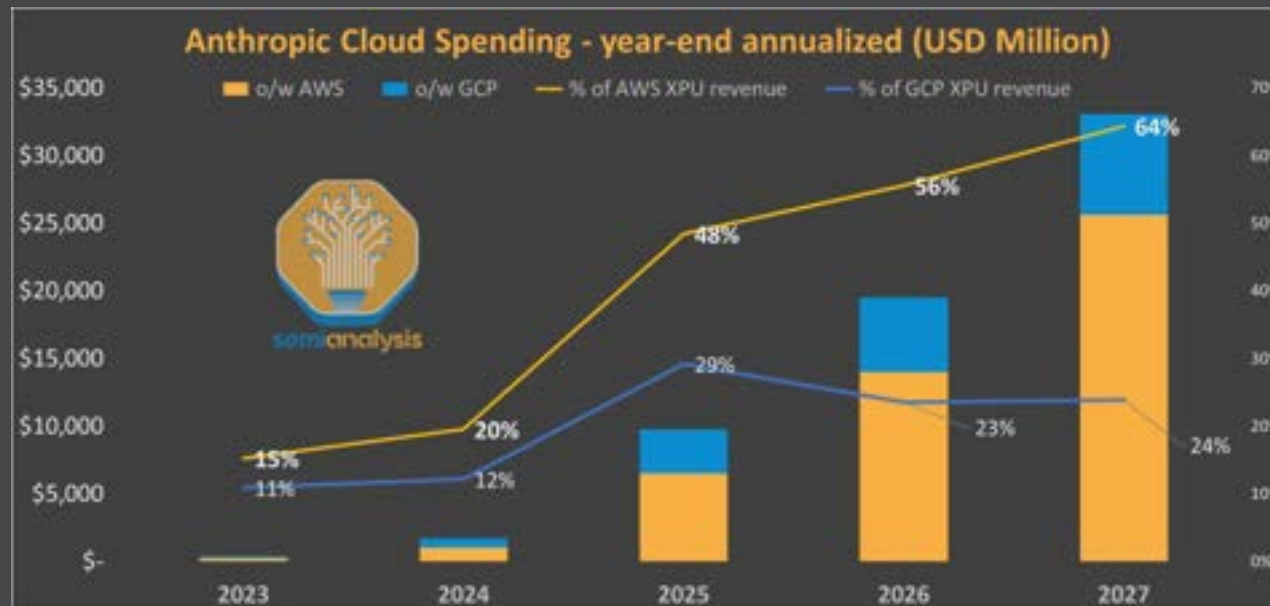
- We see higher-than-expected CapEx driven by finance-lease ROU assets additions. A few giant leased datacenters will begin operations in the next 12-18 months
- The OpenAI ramps will provide strong revenue growth in 2026.
- The loss of OpenAI is set to cause a slowdown in 2027. All forward-looking datacenter indicators are trending down.



Source: sales@semianalysis.com

Anthropic-driven AWS acceleration

- Anthropic set to weigh over 50% of the AWS XPU revenue by 2026
- AWS is overexposed to Anthropic training, not inference:
 - GCP is mostly capturing Anthropic revenue surge (i.e. inference cloud spend)
 - AWS is building GW-scale facilities for Anthropic that **aren't generating revenue yet but will turn on in Q3/Q4 2025**



Source: sales@semianalysis.com



AWS / Anthropic AI Training Datacenters as of **Q3 2025**
Source: SemiAnalysis Datacenter Industry Model



AWS / Anthropic AI Training Datacenters as of **Q3 2024**
Source: SemiAnalysis Datacenter Industry Model

sales@semianalysis.com

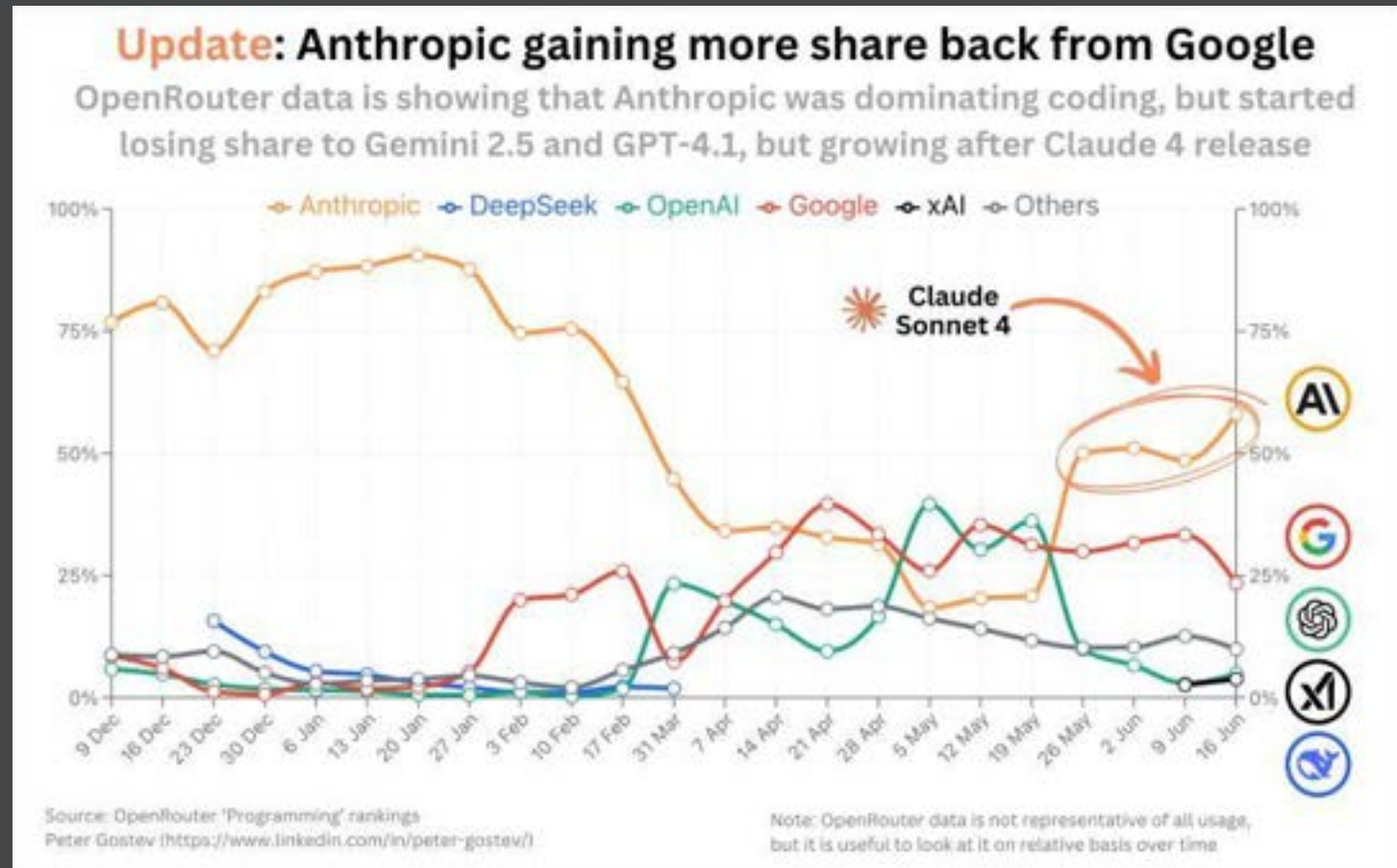
GOOGLE

GCP: OpenAI + Anthropic driving ~40% of GPU/XPU Revenue

- Google Cloud seeing good traction with AI labs and other startups

DeepMind: Gemini share gains

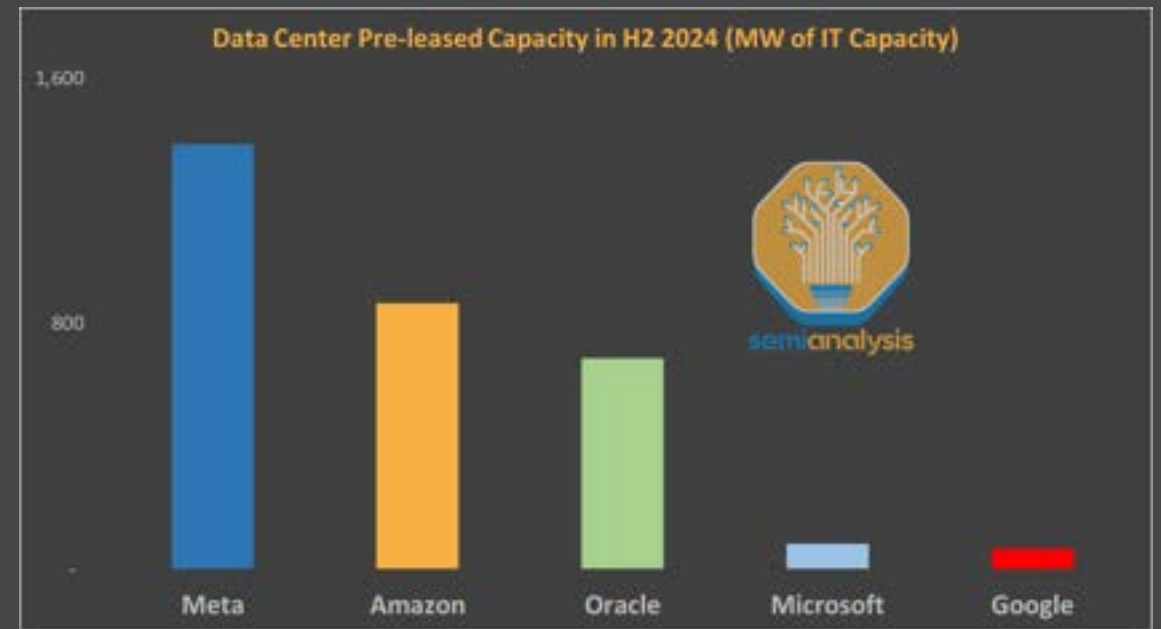
- API revenue trending well YTD (albeit recently losing momentum)
- Google is still in an “existential threat” moment – all-in on the AI race



Source: OpenRouter, Peter Gostev

CoreAI strength, LLaMa fighting back

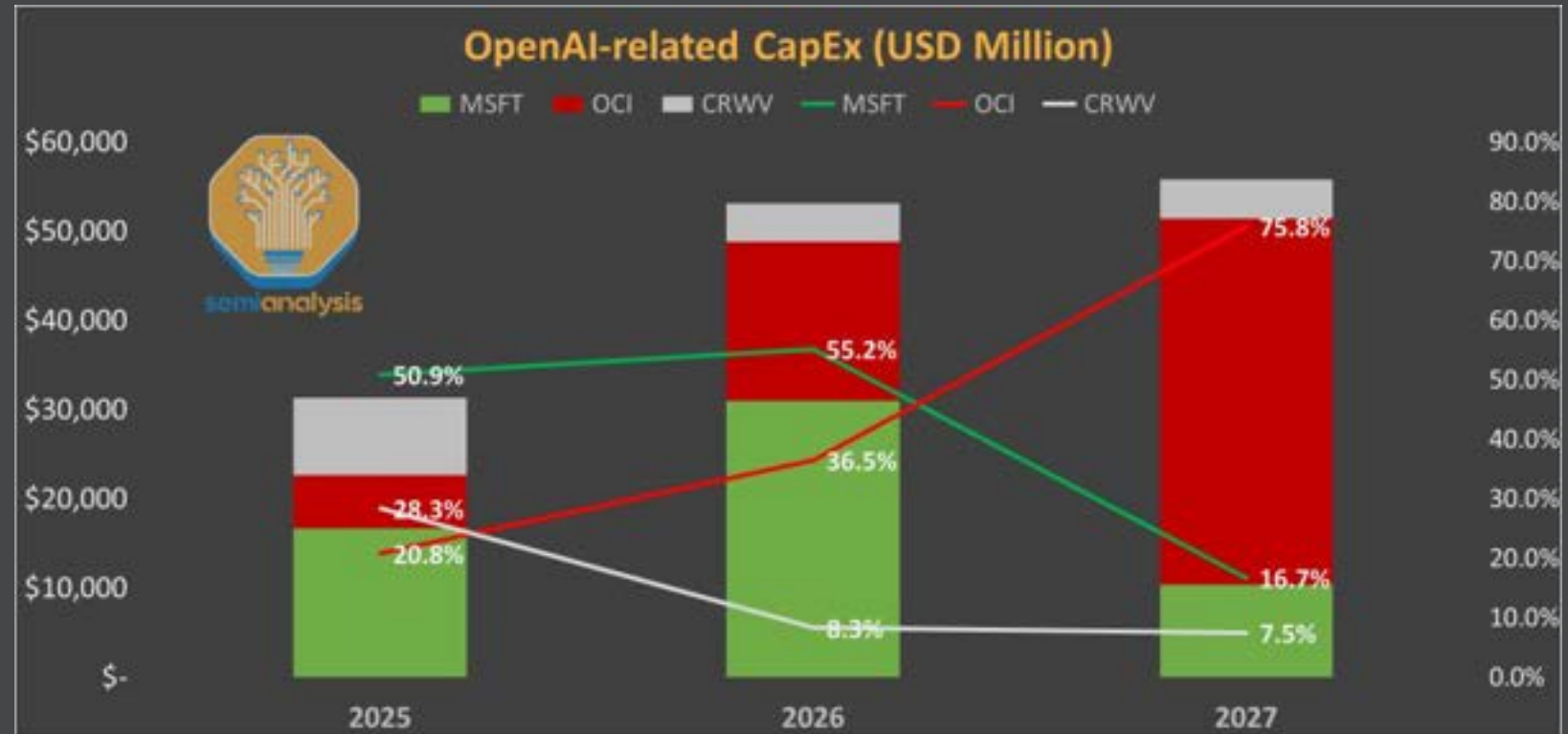
- Zuck is committing tens of billions of dollars of capital per year on LLaMa, despite no revenue yet
- Forward-looking datacenter indicators are flashing green



Source: sales@semianalysis.com

100% CapEx to sales in 2027?

- Oracle will take the majority of OpenAI-related CapEx in 2027
- Funding is the uncertainty: debt? JV (Stargate) ? Equity raise?



Source: sales@semianalysis.com

Higher-than-expected GPU margins?

- Our project-based analysis suggests a \$2.6/GPU/hr GB200 contract yields >40% operating margins
- This might be overstated due to Oracle's 6-year depreciation policy on servers

	Unit	Year 1 Mar-26	Year 2 Mar-27	Year 3 Mar-28	Year 4 Mar-29	Year 5 Mar-30
Income Statement						
Chips in Service	# Chips (GB200)	400,000	400,000	400,000	400,000	400,000
Compute Rental	USD/hr/GPU	2.60	2.60	2.60	2.60	2.60
Revenue	USD Million	9,160	9,110	9,135	9,110	8,693
Hosting Cost	USD Million	(1,001)	(1,031)	(1,062)	(1,094)	(1,127)
Electricity Cost	USD Million	(755)	(766)	(784)	(797)	(813)
Gross Profit	USD Million	7,404	7,313	7,290	7,219	6,753
Server Depreciation	USD Million	(3,295)	(3,286)	(3,295)	(3,286)	(3,286)
Amortization of Installation/Fit Out cost	USD Million	(2)	(2)	(2)	0	0
Repair and Maintenance	USD Million	(2)	(2)	(2)	(2)	(2)
Sales and Marketing Cost	USD Million	(46)	(46)	(46)	(46)	(43)
Annual Maintenance Cost	USD Million	0	(1)	(1)	(1)	(1)
Operating Profit	USD Million	4,059	3,977	3,944	3,885	3,421
<i>Operating Margin</i>	<i>%</i>	<i>44%</i>	<i>44%</i>	<i>43%</i>	<i>43%</i>	<i>39%</i>
Interest Expense	USD Million	(890)	(887)	(890)	(887)	(445)
Profit Before Tax	USD Million	3,169	3,089	3,054	2,997	2,976
Income Tax Expense	USD Million	(634)	(618)	(611)	(599)	(595)
Contribution Profit	USD Million	2,535	2,471	2,443	2,398	2,381

Source: sales@semianalysis.com

August 2025

ECONOMICS AND ROI OF GPU CLOUDS

THE AI TOKEN FACTORY ECONOMICS STACK

The AI Token Factory Economics Stack						
Layer		Units	H200	Analysis Methodology	SemiAnalysis Team	
Datacenters	Colocation Cost	USD/kW/mth	\$130	DC Market Surveys	AI Datacenters	
	Electricity Cost	USD/kWh	\$0.087	Power Grid/Market Statistics		
Neoclouds and Hyperscalers	Capital Cost per GPU	USD/GPU	\$36,620	Semicon Manufacturing Analysis	IC Design, Networking, Systems	
	Cost of Capital	%, per annum	13.3%	Neocloud Market Surveys	AI Neoclouds	
	Total Cost of Ownership	USD/hr/GPU	\$1.75			
	Neocloud Pre-tax Profit Margin	%	12.7%			
	Neocloud Rental Price	USD/hr/GPU	\$2.00			
Inference Providers	Inference Throughput/GPU ¹	Tokens/s/GPU	313	Llama 405B FP8 Benchmarking Runs	AI Engineering	
	GPU Rental Cost	USD/hr/GPU	\$2.00	Neocloud Market Surveys		
	Effective Utilization Rate	%	75%	Systems Analysis		
	Inference Compute Cost	USD/M Tok	\$2.37			
	Inference Provider Gross Margin	%	13.9%			
	Token Selling Price	USD/M Tok	\$2.75			
Applications	Tokens/User/Month	Tok/user/mth	6,000,000	Inference market research, surveys, proprietary dataset analysis		AI Tokenomics
	Token Unit Cost	USD/M Tok	\$2.75			
	Monthly Active Users	M Users	1.8			
	Average Revenue per User	USD/mth	\$25.00			
	Gross profit per User/Mth	USD/mth	\$8.50			
	Application Gross Margin	%	34.0%			



seminalysis

1. Llama 405B FP8 on H200 using VLLM. Uses 1k input, 4k output tokens, ~190 seconds end to end latency.

1. Llama 405B FP8 on H200 using VLLM. Uses 1k input, 4k output tokens, ~190 seconds end to end latency.

Source: sales@semianalysis.com

TOTAL COST OF OWNERSHIP FOR H200

TCO Summary for H200		
	Unit	H200 SXM
<u>Cluster Capital Costs</u>		
GPU Provider Content (per GPU)	USD	
GPU Provider Content + Warranty (per Server)	USD	
Other Server Costs	USD	
Server Cost	USD	\$212,579
Service, Networking, Storage, Software, Others	USD	\$80,377
Total Upfront Cluster Capex, per Server	USD	\$292,956
Total Upfront Cluster Capex, per Logical GPU	USD	\$36,620
Weighted Average Cost of Capital	%	13.3%
Useful Life in Years	Years	4
Total Cluster Capital Costs per Month per Server	USD/mth	\$7,896
Capital Cost per Unit, per Hour	USD/hr/GPU	\$1.35
Operating Cost per Unit, per Hour	USD/hr/GPU	\$0.39
Total Cost per Unit per Hour	USD/hr/GPU	\$1.75

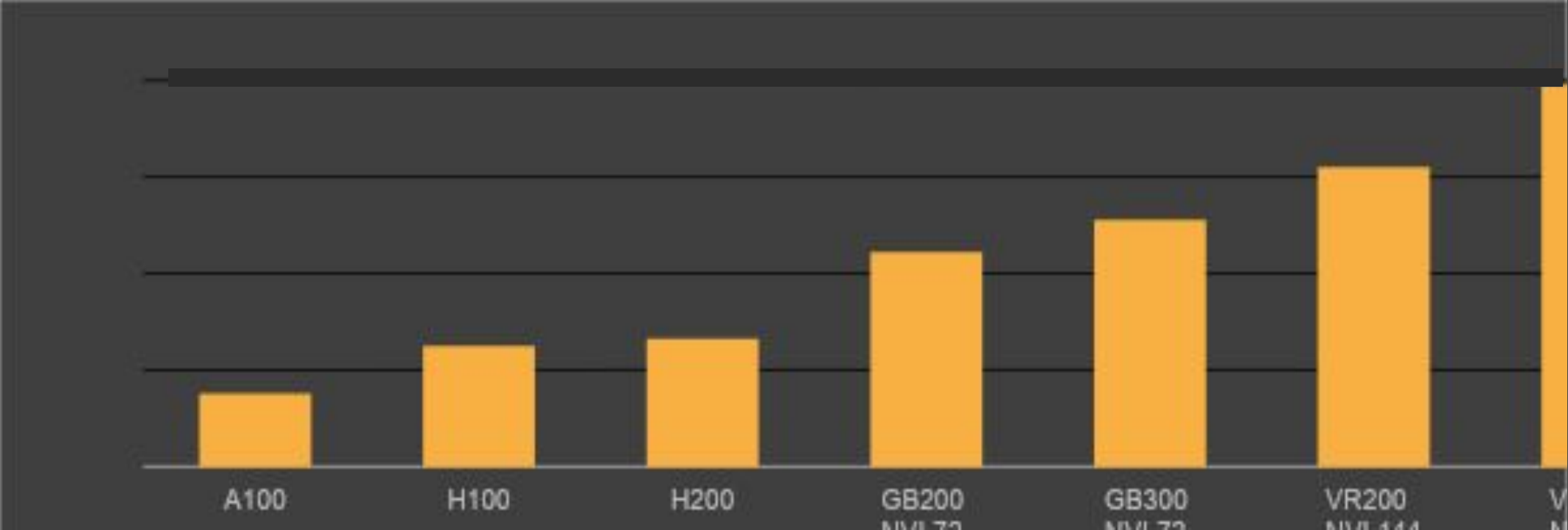
Source: sales@semianalysis.com

TOTAL COST OF OWNERSHIP FOR H200

AI Cloud Operating Cost of Ownership		
	Unit	H200 SXM
<u>Cluster Operating Costs</u>		
Electricity Cost	USD/kWh	\$0.0870
Utilization Rate	%	80%
Power Usage Effectiveness (PUE)	Ratio	1.35
Electricity Cost per kW of Critical IT per mth	USD/kW/mth	\$68.6
Colocation Cost	USD/kW/mth	\$130.0
Total Hosting Cost per kW per Month	USD/kW/mth	\$198.6
All-in Power Consumption	kW	10.72
Total Server Hosting Costs per Month	USD/mth	\$2,128
Remote Hands + Support Engineer	USD/mth	\$131
Internet Connection	USD/mth	\$39
Total Cluster Operating Cost per Month, per Server	USD/mth	\$2,298
Total Cluster Operating Cost per Month, per GPU	USD/mth	\$287
<i>per Hour per GPU</i>	<i>USD/hr/GPU</i>	<i>\$0.39</i>

Source: sales@semianalysis.com

AI ACCELERATOR ROADMAP



	A100	H100	H200	GB200 NVL72	GB300 NVL72	VR200 NVL144	VR300 NVL576
Memory Capacity (GB)	80	80	141	192	288	288	1,024
Memory BW (TB/s)	2.0	3.4	4.8	8.0	8.0	13.0	32.0
Logic Die Configuration	1 x Reticle Sized GPU			2 x Reticle Sized GPU		2 x Reticle GPU, 2 x I/O	4 x Reticle GPU, 4 x I/O
Interposer Size (# Reticles)	1.9x	1.9x	1.9x	3.3x	3.3x	4.8x	9.6x
TDP (W)	400	700	700	1,200	1,400	1,800	2,800
Dense Flops FP8 (TFLOPs)	312	1,979	1,979	5,000	5,000	16,667	33,333

Source: sales@semianalysis.com

August 2025

SEMIANALYSIS INDUSTRY MODELS

AI ACCELERATOR MODEL

A Comprehensive Model of supply/demand dynamics in the AI Accelerator Supply Chain

- Analyzes chip production capacity, shipments and install base by triangulating datapoints across the semiconductor production and design stack.
- Our Market Intelligence is trusted by top US hyperscalers, many global semiconductor companies, leading international investment firms that invest in both private and public markets, as well as various chip design and AI startups.

The AI Accelerator Model includes the following datapoints:

Number of shipments and ASPs of AI accelerators by SKU. Covered product lines include:

- Nvidia A100, H100, B100, B200 (1000W, 1200W, NVL72), X/R100, Y100, Z100,
- Google TPU v4, v4i, v5p, v5e, v6p, v6e, v7, v7e,
- Meta MTIA Gen 2, MTIA Gen 3,
- AWS Inferentia2, Inferentia3, Inferentia4, Trainium1, Trainium2, and Trainium3,
- Microsoft Maia Athena and Braga,
- AMD MI300, MI400,
- Intel Habana Gaudi2 and Gaudi3, Falcon Shores

Supply chain and capacity orders for these chips which includes:

- Number of Foundry wafers, Number of 2.5D volumes and yields (TSMC, Intel, Amkor, ASE)
- Total number of die attach steps (BESI, ASMPT, etc)
- HBM: type, capacity layer count, number of stacks, bits, manufacturer

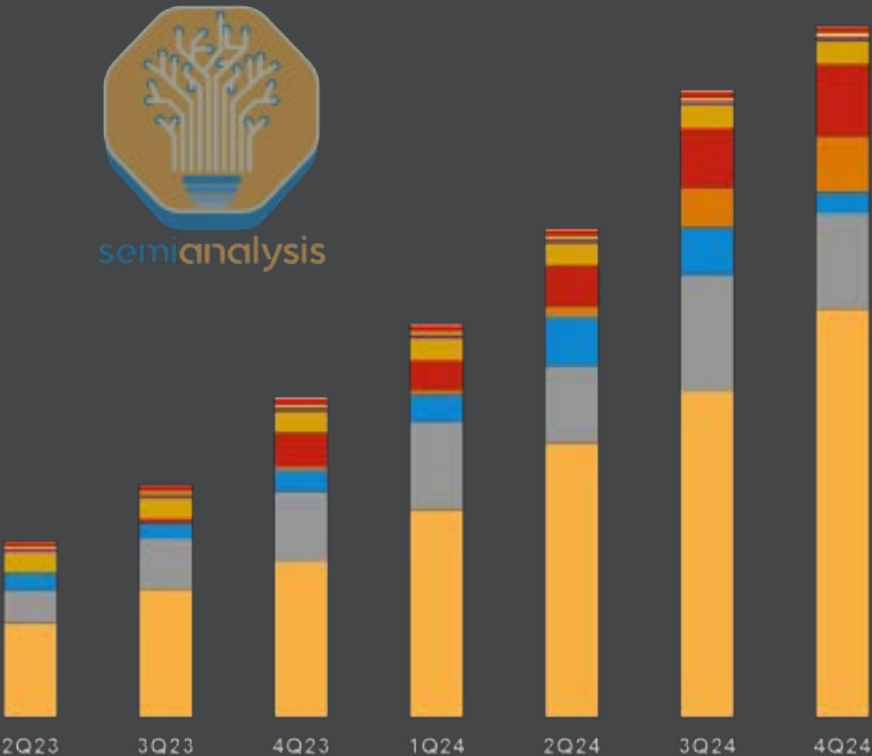
AI Accelerator Shipments and Installed base by customer:

- US Hyperscalers (Microsoft/OpenAI, Google, Meta, Amazon, Oracle)
- Chinese Hyperscalers (Baidu, Alibaba, Tencent, Bytedance)
- Neoclouds (Coreweave, Applied Digital, Crusoe, SF Compute, Genesis Cloud, and 10s more)
- Enterprises: Tesla/x.AI,
- Sovereign Initiatives: Singapore, Canada, Qatar, and more



SEMIANALYSIS PRODUCT OVERVIEW

CoWoS Quarterly Output



Nvidia vs Total Accelerator Market



DATACENTER INDUSTRY MODEL

An Integrated and Holistic Datacenter Supply/Demand Analysis

- Integrates supply chain and production analysis from the AI Accelerator Model with extensive surveillance of existing datacenter capacity and explanation of future expansion.
- Tracks current and future estimated install base (and thus AI Datacenter Critical IT Power demand) for Hyperscalers, AI Neoclouds, Sovereign AI and National Labs and more.
- Identifies likely supply shortfalls by hyperscale provider and on a geographical basis – by region, power grid, or data center market.

The AI Datacenter Model includes the following data points:

- Accelerator Unit & Critical IT Power Forecasts:
 - Shipments, deployments, and AI Data Center Requirements through 2030
 - Power needs per unit across major chips/models
 - AI data center power demand by Geography
- Datacenter Capacity & Supply Analysis:
 - Current and forecast capacity by provider and region
 - Industry wide capex breakdown
 - AI Supply/ Demand analysis by hyperscaler and region
- Power Considerations:
 - AI vs non-AI data center power demand
 - Training / inference power and emissions for LLM models
 - Global electricity tariffs and solar economics



INDUSTRY MODEL – AI DATACENTER MODEL

Integrated and Holistic Datacenter Supply/Demand Analysis

- **DEMAND:** how much power will AI Accelerator deployments in 2023-2030 require? Broken down by company (70+ individual buyers of AI chips)
- **SUPPLY:** is the industry overbuilding, underbuilding? Broken down by region and company – tracking 5,000+ data centers globally (ex-China)
- **Methodology:** Real-time satellites, building permits, ISO/RTO substation & transmission data, channel checks via 10+ datacenter conferences attended every year



INDUSTRY MODEL – AI DATACENTER MODEL

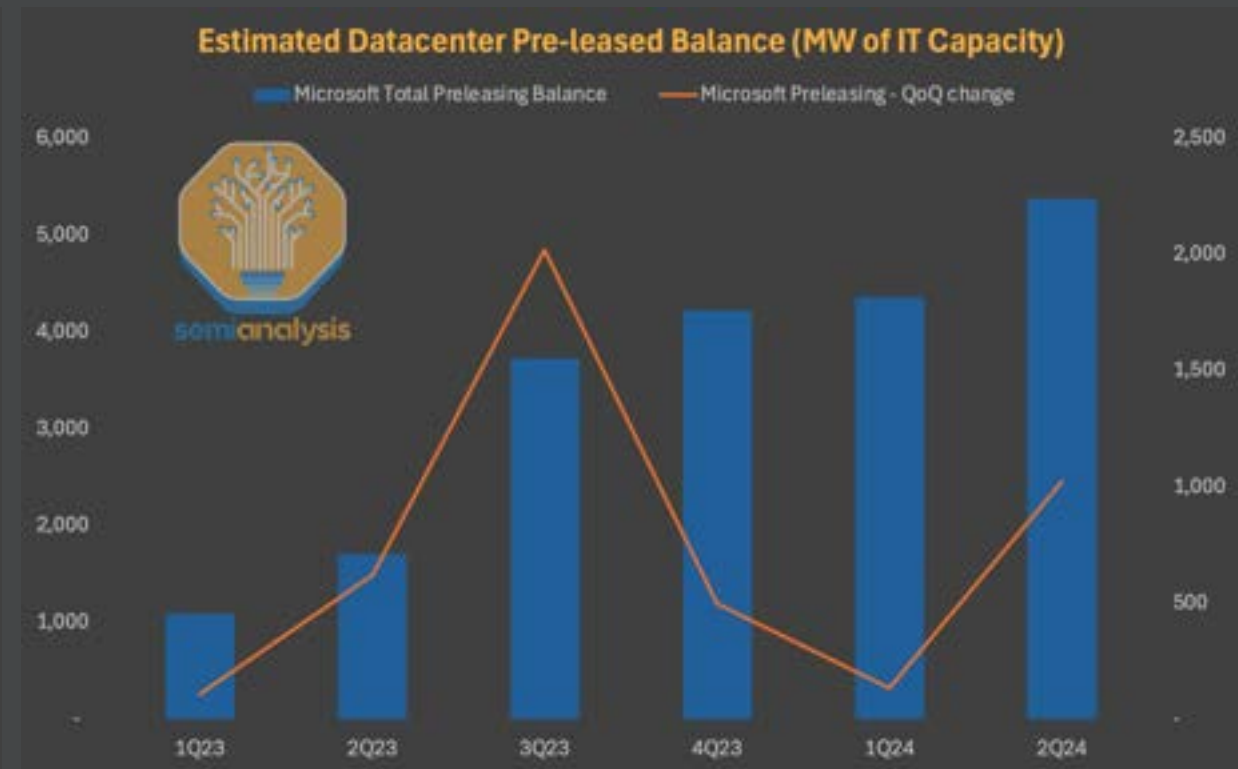
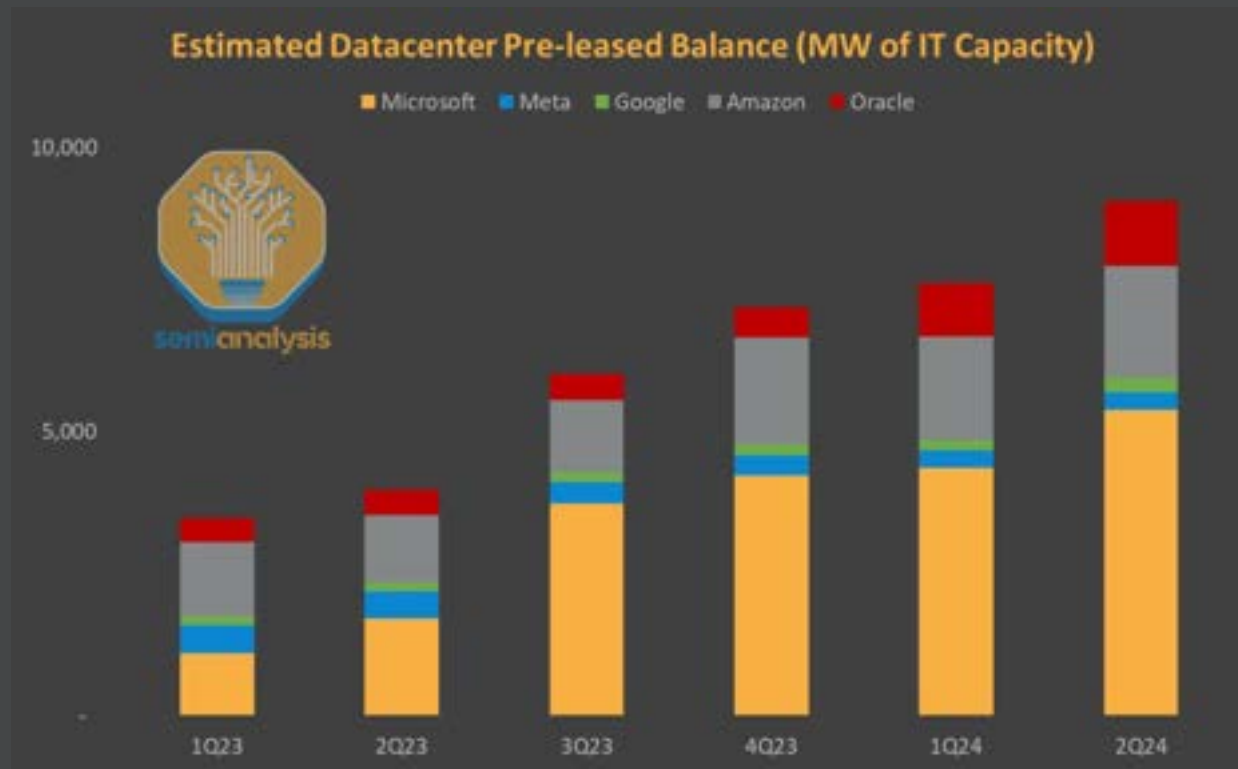
KEY INSIGHTS:

- **Hyperscaler CapEx leading indicator**
 - Tracking leasing & self-build plans indicates 12mo+ growth plans
 - Identifying capacity shortfalls and likely outcomes, e.g. a constrained hyperscaler likely to contract more Datacenter or GPU Cloud capacity
 - Cloud Computing growth driven by capacity delivery timeline
- **Datacenter Load Growth by Geography:**
 - Current and forecast capacity by provider and region
 - Industry wide capex breakdown
 - AI Supply/ Demand analysis by hyperscaler and region
- **AI training and inference plans:**
 - Identifying largest planned AI clusters
 - Understanding OpenAI, Anthropic, xAI etc. capacity growth



INDUSTRY MODEL – AI DATACENTER MODEL

Tracking hyperscaler self-build, leasing, pre-leasing



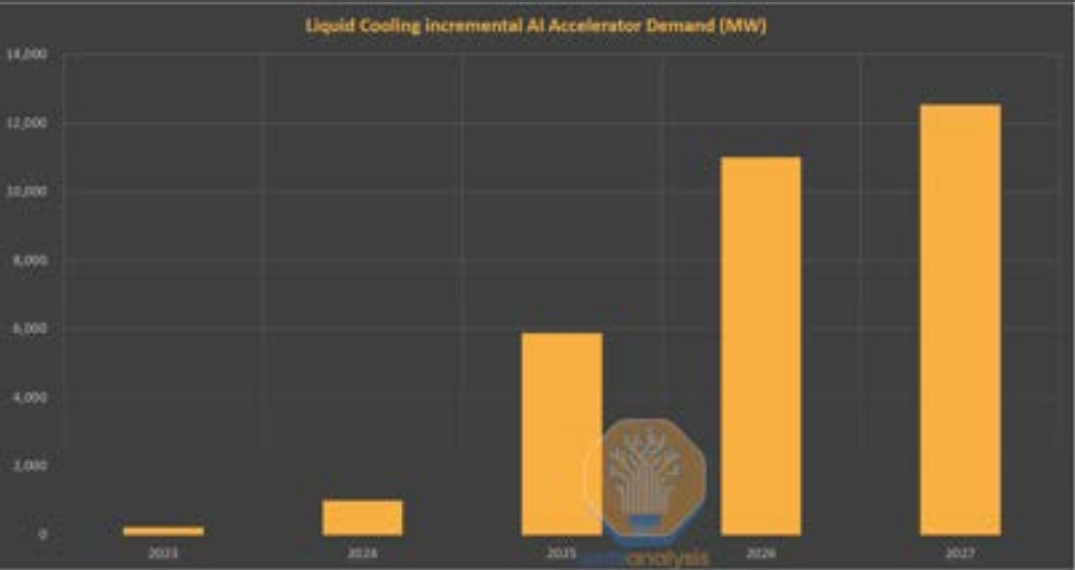
INDUSTRY MODEL – AI DATACENTER MODEL

Identifying construction pauses and slowdowns before public announcements



INDUSTRY MODEL – AI DATACENTER MODEL

Unique insights into power & cooling for 80+ AI Accelerators



Nvidia Roadmap				
	2024	2025	2026	2027
AI Accelerator				
	Blackwell		Rubin	
	GB200 NVL36/72	GB300 NVL72	R100	Rubin Ultra
Peak Rack Density in kW	132	163	303	819
% of Volumes on Rack-scale format	67%	69%		
Total MWs shipped with the highest density SKU				

March 2025

SEMIANALYSIS CORE RESEARCH

ADDITIONAL SLIDES

CORE RESEARCH PRODUCT OVERVIEW

LEVERAGING SEMIANALYSIS RESEARCH FOR ACTIONABLE INSIGHTS INTO PUBLIC EQUITIES

- Focused research on **technology inflections** & **financial implications**
- **Product Roadmap**
 - **Weekly Databook:** Industry, equity, and alt data driving investment decisions
 - **Thematic Notes:** Recurring coverage of key themes including the semi-cycle
 - **Product Notes:** Company product launches and financial impact
 - **Conference Notes:** Summaries of key presentations at industry conferences
 - **Trading Commentary:** Earnings and other catalyst events
- **Upgraded tier product offerings**
 - **Models:** Select company and product models and research
 - **Calls:** Real-time Q&A and discussion

TECHNOLOGY AND PRODUCT NOTES

FINANCIAL IMPLICATIONS OF PRODUCTS AND TECHNOLOGIES

- Ad hoc notes detailing content gains and losses for component suppliers
- In-depth overview of technologies and products and their impacts on companies
- Examples
 - GB300 NVL36x2 Cancellation – 10/2/24 (shown on the right)
 - Quantified dollar content loss for SMTC, MTSI, APH, and TEL
 - NVDA VRM Content – 10/25/24
 - Detailed MPWR share loss prior to MPWR earnings
 - Astera Labs PCIe Switch – 10/8/24
 - Product overview, expected adoption, and attach rates

Excerpt from Astera Labs Update | ASPs, Volumes, Further Switch Detail February 12, 2025

We have talked about Astera Labs' content in Amazon Trn2 servers and the revenue opportunity from its PCIe switch portfolio. We update our views in this note, including the benefit of an ASP increase for the paddle cards it sells for Trn2 Ultra servers and content for Trn2e servers. This analysis also includes the impact of the recent increase to unit estimates from the latest **Accelerator Model Update**. Our work builds off the Accelerator Model and focuses primarily on Astera's retimer business and recently launched switch product. While we assume Street estimates for Taurus and Leo, we see upside for the Taurus AEC business, especially in 2H25, given AEC content in Trn2 and GB200.

Key Takeaway: Street revenue estimates for 2025 are slightly below our ALAB revenue forecast despite our above consensus NVIDIA unit estimates. This is likely a result of our view of lower retimer attach per GPU in GB systems. However, we are above Street numbers for FY26 and FY27 driven by the Switches ramp.

ALAB Revenue Estimates (\$ in thousands)					
	2024	2025	2026	2027	
Retimer Revenue (Artes)	\$ 339,132	\$ 430,202	\$ 575,004	\$ 528,700	
PCIe Switch Revenue	-	\$ 54,601	\$ 172,377	\$ 281,600	
LANlink Switch Revenue	-	-	\$ 35,750	\$ 290,000	
Switch Revenue (Scorpio)	-	\$ 54,601	\$ 227,627	\$ 681,600	
Other Revenue (Taurus + Leo)	\$ 47,047	\$ 100,313	\$ 189,582	\$ 299,008	
Total Revenue	\$ 386,179	\$ 585,118	\$ 982,223	\$ 1,509,308	
Street	\$ 381,077	\$ 615,308	\$ 771,403	\$ 1,194,000	
% vs. Street	0.8%	-4.9%	18.9%	82.9%	

Source: SemiAnalysis, Bloomberg

ASP Update

AWS Paddle Cards

As a reminder, Astera provides paddle cards in addition to standard package retimers for Trainium2 Ultra racks. These paddle cards contain the retimer, PCB, power delivery, and other components; therefore, it carries a higher ASP than standard package retimers. We now believe Astera will earn \$70 per paddle card for Trn2 Ultra racks, significantly higher than our previous estimate of a \$50 ASP. Trn2e Ultra racks will get another ASP uplift to \$80, as the 2e uses a 5nm paddle card compared to the 12nm paddle card in the Trn2. The more advanced retimer will result in 20%-30% lower power, which will allow the accelerators to run at higher power. Astera sells two retimers for every GPU in Ultra racks, so these ASP uplifts have a material top line impact.

AWS Standard Package Retimers

Trn2 racks utilize Gen 5 16nm retimers that carry an ASP of ~\$35. We believe Trn2

CONFERENCE NOTES

MAIN TAKEAWAYS FROM INDUSTRY AND COMPANY CONFERENCES

- Conference schedule and notes
- Conference summary
 - Major theme / story of the conference
 - **Technical checks:** confirm or deny technologies and trends
 - **Financial checks:** confirm or deny sockets and attach rates
 - Biggest changes in technology themes from prior year
 - Most exciting and disappointing technologies

*Excerpt from the
Conference Note on PTC 2025*

Conference Takeaways

We met with over 20 datacenter colocation operators, fund management platforms and equipment vendors at the Pacific Telecommunications Council (PTC) Conference. The conference was very telecom focused in the past, but now we think attendees are split 50:50 between telecom and datacenter. It is a very 1:1 meeting focused conference, where datacenter and telecom operators meet with clients and potential partners. We outline a few key themes discussed at the conference:

- **Hyperscalers vs Neoclouds:** Understanding of the Neocloud business model is still very nascent, and very few have made any serious deployments outside of the US. Most are not aware of key operators. **Colocation operators are still very focused on Hyperscaler deployment** given strong credit profiles and good 7-10y colocation terms. A few datacenter operators are interested in exploring the Neocloud space, however.
- **GPU Diffusion Regulations:** Almost all DC operators are still trying to digest the regulations and have not yet had discussions with their clients on potential impacts. Operators with significant assets in Malaysia, India, and Latin America are particularly concerned, but are still processing the regs and many don't understand certain nuances of the rules. Indian operators are less concerned than Malaysia or LatAm, as GPU deployments in India will be hyperscale centric, with minimal Neocloud or Oracle/Bytedance complex deployments given geopolitics between India and China.
- **GPU Deployments Globally:** GPU deployments outside of the US are still relatively small. Coreweave has reportedly deployed clusters in Europe and a few Neocloud Giants have medium-sized deployments (c. 10-30k GPUs) in the Nordics. Turning to Asia, **in Johor, Malaysia liquid cooled GPU clusters should be shipping in the first half of the year** (assuming regulations do not derail plans there), though a sizeable portion are to be deployed via Oracle, who will need to ensure they obtain the NVEU exception. In India, deployments are still very nascent – we estimate fewer than 10,000 GPUs deployed in proper clusters at various Neoclouds in the country. US Hyperscale deployments will be the main drivers of GPU deployments in India for the next few years given the low total GPU country cap. Outside of this, most major European and Asian countries have one or two Neoclouds with a few thousand H100 equivalent GPUs.
- **Reliance Industries AI DC Project a Keep in View:** Many DC operators in India have made large capex announcements, of which **Reliance Industries' 3GW+ datacenter is the most notable**. This project, first announced by Mukesh Ambani on stage at the Nvidia India Conference in October 2024, has yet to break ground, but it is important to follow as Reliance has abundant access to capital and has a proven track record of successfully entering many capital-intensive industries.

RECURRING AND AD HOC ANALYSES FOCUSED ON EQUITY MARKETS

- Weekly report updating key industry KPIs and developments
- Recurring third party and proprietary dataset updates
 - Export/import data, monthly Taiwan revenue, GPU pricing, etc.
- Earnings
 - Focus on key read-throughs and end market implications
- Trading commentary
 - Drivers of industry and stock performance

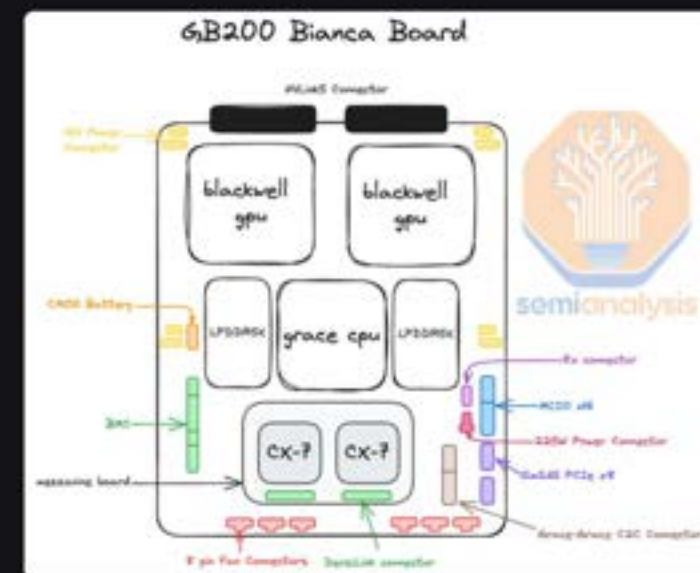
Supply Chain Update

For some quick background, the Hopper 50M module comes directly from NVIDIA. NVIDIA chooses the component supplier, and that company gets 100% share. So, Monolithic Power (MPWR) got 100% share on Hopper until Renesas came into play. TXN will also be getting content, and Infineon will receive some share later. **We do not believe VECR has any content.** We previously wrote about the [GB200 hardware architecture](#) and VECR's content loss at NVIDIA in our note on [power delivery in AI accelerators](#).

For the B200 and B300A, the supply chain remains the same and MPWR will lose share as these newer players gain incremental share. Given volume for these platforms is down more than power content is up, this is generally incrementally negative.

The price per SPS has fallen from \$1.35 to ~\$0.60 over the last 2 years.

However, there is something bigger at play with the shift to the Bianca board. The Bianca board, just like the HGX modules and USB board, has a bunch of VRMs. However, NVDA will only directly purchase the 12V to 0.8V VRM for the Bianca board.



Source: SemiAnalysis

COMPLIANCE

SemiAnalysis has adopted a comprehensive compliance program by engaging an experienced third-party compliance firm. We are not a FINRA member, but we have adopted research practices that are up to FINRA standards.

- Compliance policies adopted are outlined in an extensive compliance handbook which all staff must review annually and upon onboarding.
- We conduct compliance training during onboarding and annually, covering applicable laws and regulations, proper handling of material non-public information (MNPI), regulatory risk, personal account investments and trading, business communications, cybersecurity among other topics.
- We have implemented personal account monitoring, and we restrict trading on securities related to the Semiconductor and AI sector (a list of hundreds of securities) as well as on any other securities that feature prominently in articles we plan to publish or have published recently.
- Institutional research for institutional investors are reviewed by a Series 16 Supervisory Analyst prior to publication.
- Client communications are subject to compliance policies, and we have implemented appropriate monitoring and record-keeping systems.
- We do not provide any investment advice, buy/sell/hold or any other ratings on any securities, nor do we provide valuations or target prices for any securities. We provide what we believe is the most realistic view into how the industry and firms do from a revenue, cost, and margin perspective by product or segment on a historical and forward-looking basis.

DISCLAIMERS

Analyst Certifications and Independence of Research.

Each of the analysts whose names appear in this report hereby certify that all the views expressed in this Report accurately reflect our personal views about any and all of the subject securities or issuers and that no part of our compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views of in this Report.

SemiAnalysis LLC (the “Company”) is an independent equity research provider. The Company is not a member of the FINRA or the SIPC and is not a registered broker dealer or investment adviser. SemiAnalysis has no other regulated or unregulated business activities which conflict with its provision of independent research.

Limitation Of Research And Information.

This Report has been prepared for distribution to only qualified institutional or professional clients of SemiAnalysis LLC. The contents of this Report represent the views, opinions, and analyses of its authors. The information contained herein does not constitute financial, legal, tax or any other advice. All third-party data presented herein were obtained from publicly available sources which are believed to be reliable; however, the Company makes no warranty, express or implied, concerning the accuracy or completeness of such information. In no event shall the Company be responsible or liable for the correctness of, or update to, any such material or for any damage or lost opportunities resulting from use of this data.

Nothing contained in this Report or any distribution by the Company should be construed as any offer to sell, or any solicitation of an offer to buy, any security or investment. Any research or other material received should not be construed as individualized investment advice. Investment decisions should be made as part of an overall portfolio strategy and you should consult with a professional financial advisor, legal and tax advisor prior to making any investment decision. SemiAnalysis LLC shall not be liable for any direct or indirect, incidental or consequential loss or damage (including loss of profits, revenue or goodwill) arising from any investment decisions based on information or research obtained from SemiAnalysis LLC.

Reproduction and Distribution Strictly Prohibited.

No user of this Report may reproduce, modify, copy, distribute, sell, resell, transmit, transfer, license, assign or publish the Report itself or any information contained therein. Notwithstanding the foregoing, clients with access to working models are permitted to alter or modify the information contained therein, provided that it is solely for such client's own use. This Report is not intended to be available or distributed for any purpose that would be deemed unlawful or otherwise prohibited by any local, state, national or international laws or regulations or would otherwise subject the Company to registration or regulation of any kind within such jurisdiction.

Copyrights, Trademarks, Intellectual Property.

SemiAnalysis LLC, and any logos or marks included in this Report are proprietary materials. The use of such terms and logos and marks without the express written consent of SemiAnalysis LLC is strictly prohibited. The copyright in the pages or in the screens of the Report, and in the information and material therein, is proprietary material owned by SemiAnalysis LLC unless otherwise indicated. The unauthorized use of any material on this Report may violate numerous statutes, regulations and laws, including, but not limited to, copyright, trademark, trade secret or patent laws.

Q&A

Reach Out To Analysts

Bloomberg Intelligence

- Woo Jin Ho: who88@bloomberg.net
- Jake Silverman: jsilverma109@bloomberg.net

SemiAnalysis

- Jeremie Ontiveros jeremie@semianalysis.com
- Sales sales@semianalysis.com

Disclaimer

The data included in these materials are for illustrative purposes only. The BLOOMBERG TERMINAL service and Bloomberg data products (the “Services”) are owned and distributed by Bloomberg Finance L.P. (“BFLP”) except (i) in Argentina, Australia and certain jurisdictions in the Pacific islands, Bermuda, China, India, Japan, Korea and New Zealand, where Bloomberg L.P. and its subsidiaries (“BLP”) distribute these products, and (ii) in Singapore and the jurisdictions serviced by Bloomberg’s Singapore office, where a subsidiary of BFLP distributes these products. BLP provides BFLP and its subsidiaries with global marketing and operational support and service. Certain features, functions, products and services are available only to sophisticated investors and only where permitted. BFLP, BLP and their affiliates do not guarantee the accuracy of prices or other information in the Services. Nothing in the Services shall constitute or be construed as an offering of financial instruments by BFLP, BLP or their affiliates, or as investment advice or recommendations by BFLP, BLP or their affiliates of an investment strategy or whether or not to “buy,” “sell,” or “hold” an investment. Information available via the Services should not be considered as information sufficient upon which to base an investment decision. The following are trademarks and service marks of BFLP, a Delaware limited partnership, or its subsidiaries: BLOOMBERG, BLOOMBERG ANYWHERE, BLOOMBERG MARKETS, BLOOMBERG NEWS, BLOOMBERG PROFESSIONAL, BLOOMBERG TERMINAL and BLOOMBERG.COM. Absence of any trademark or service mark from this list does not waive Bloomberg’s intellectual property rights in that name, mark or logo. All rights reserved. © 2025 Bloomberg.

Bloomberg Intelligence is a service provided by Bloomberg Finance L.P. and its affiliates. (“Bloomberg”). Bloomberg is not an officially recognized credit rating agency in any jurisdiction, and customers should not use or rely on Bloomberg Intelligence to comply with applicable laws or regulations that prescribe the use of ratings issued by accredited or otherwise recognized credit rating agencies. Bloomberg Intelligence Credit and Company research may not be available in certain jurisdictions.

Bloomberg Intelligence shall not constitute, nor be construed as, investment advice or investment recommendations (i.e., recommendations as to whether or not to “buy,” “sell,” “hold,” or to enter or not to enter into any other transaction involving any specific interest) or a recommendation as to an investment or other strategy. No aspect of the Bloomberg Intelligence function is based on the consideration of a customer’s individual circumstances. Bloomberg Intelligence should not be considered as information sufficient upon which to base an investment decision. Customers should determine on their own whether they agree with Bloomberg Intelligence. Bloomberg Intelligence should not be construed as tax or accounting advice or as a service designed to facilitate any Bloomberg Intelligence customer’s compliance with its tax, accounting, or other legal obligations. Bloomberg believes that the information it uses in Bloomberg Intelligence comes from reliable sources, but does not guarantee the accuracy of information contained in Bloomberg Intelligence. Employees involved in Bloomberg Intelligence may hold positions in the securities analyzed or discussed on Bloomberg Intelligence.



**Bloomberg
Intelligence**

Data-driven research.

350 research professionals
covering 2,000 companies
and more than 135 industries

 **BI** <GO>



This virtual event has ended.

To discover more Bloomberg webinars,
please visit [bloomberg.com/webinars](https://www.bloomberg.com/webinars)